



Prometheus

1. WHAT IS PROMETHEUS?

Prometheus is a monitoring and alerting stack that was created back in 2012 at SoundCloud and has been part of the CNCF (Cloud Native Computing Foundation) since 2016.

Prometheus is able to handle monitoring for time-series metrics in its stride and is able to perform functions such as addition, averaging metrics over time and multiplying any time-series metrics it is given.

2. What Is Prometheus Monitoring?

Prometheus monitoring provides a way for users to collect and process metrics from their applications by scraping and capturing time-series metrics.

3. What Is Prometheus Monitoring Used For?

The Prometheus monitoring tool was initially created due to the lack of tools available that were suitable for monitoring dynamic cluster scheduling within containers.

4. What Is PromQL?

The dedicated querying language used by Prometheus to perform calculations and a wide array of other functions is known as PromQL.

5. Why Would You Use Prometheus & Grafana Together?

As Prometheus is able to collect time-series metrics easily and Grafana provides a solution for transforming this data into meaningful reports and data visualisation, using these two tools on top of each other offers users many benefits.

6. Is Prometheus A Time Series Database?

Prometheus is classified as an open-source time-series database. A time-series database is designed to store and retrieve time-series data along with its associated timestamps.

A commonly seen example of time-series metrics that you may have seen visualised previously is a stock price tracker.

7. How Do You Find The Number Of Kubernetes Pods Per Namespace?

To find the number of pods per namespace you will need to use the PromQL command `sum by (namespace) (kube_pod_info)`

8. How Do You Find The CPU Usage By Instance?

To find the CPU usage by instance you will need to use the PromQL command `100 * ( 1 - avg by(instance) (irate(node_cpu{mode='idle'}[5m])))`

9. How Do You Find The HTTP Error Rate As A Percentage Of Total Traffic?

To find the HTTP error rate as a percentage of total traffic you will need to use the PromQL command `rate(http_requests_total{status_code=~"5.*"}[5m] / rate (http_requests_total[5m])`

10. How Do You Find The Increase Between Recent Samples?

To find the increase between recent samples you will need to use the PromQL command `irate(http_requests_total[10m])`

11. How Do You Display The Time Series Of A Metric?

To display the time series of a metric you will need to use the PromQL command `{_name_="node_cpu_seconds_total"}`

12. How Do You Return Everything From The Query Time Looking Back?

To return everything from the query time looking back you will need to use the PromQL command `up offset 10m`

13. How Do You Set A Bound On The Returned Values?

To set a bound on the returned values you will need to use the PromQL command `clamp_min(go_goroutines,20)`

14. How Do You Get The Total Number Of Times Compaction Happens Per Second Per Instance?

To get the total number of times compaction happens per second per instance you will need to use the PromQL command `sum by(instance)(rate(prometheus_tsdb_compactiong_duration_seconds_count[5m]))`

15. How Do You Get The Size Of The Biggest Mounted Filesystem On Each Machine?

To get the size of the biggest mounted filesystem on each machine you will need to use the PromQL command `# Using by max by (instance)(node_filesystem_size_bytes)`

16. How Do You Get Changes In Memory Usage In The Node Exporter Over The Past Hour?

To get changes in memory usage in the node exporter over the past hour you will need to use the PromQL command `process_resident_memory_bytes{job="node"} - process_resident_memory_bytes{job="node"} offset 1h`

17. How Do You Select All HTTP Status Codes Except For 4xx Ones?

To select all HTTP status codes except for 4xx ones you will need to use the PromQL command `http_requests_total{status!~"4.."}`

18. How Do You Return The K Time Series With The Highest Values?

To return the K time series with the highest values you will need to use the PromQL command `Topk`

19. How Do You Return The K Time Series With The Lowest Values?

To return the K time series with the lowest values you will need to use the PromQL command `Bottomk`

20. How Do I Measure The Total CPU Usage By The Process?

To measure the total CPU usage by the process you will need to use the PromQL command `Process_cpu_seconds_total`

21. How Do I Add A New Label Name In Prometheus?

To add a new label name you will need to use the PromQL command `Prometheus label_replace`

PROMETHEUS COMPONENTS

- **NODE-EXPORTER** :- IT IS AN PROMETHEUES AGENT , WHICH RUNS ON ALL THE SERVERS AND GATHER SYSTEM METRICS , LIKE CPU-UTILIZATION, MEMORY UTILIZATION AND ETC
- **SNMP** :- SIMPLE NETWORK MANAGEMENT PROTOCOL , USEFUL TO PULL NETWORK METRICS RELATED STATICS OR METRICS
- **PROMETHEUS SERVER**:- IN THIS SERVER THE PROMETHEUS WILL RUNN AND THEN ON ALL THE SERVER YOU WANT TO MONITOR YOU WILL TELL THIS PROMETHEUS SERVER YOU'RE TARGETS AND THIS PROMETHEUS SERVER THEN START MONITORING THESE TARGET SERVERS. AND ALL OF THIS CONFIGURATION WILL HAPPEN IN THE PROMETHUES CONFIGURATION FILE.

NOTE:- PROMETHEUS STORES CONFIGURATION IN YML FORMAT, PROMETHEUS CONFIGURATION FILE IS NAMED: prometheus.yml

PROMETHEUS ARCHITECTURE

NODE EXPORTER, IT IS INSTALL ON ALL THE SERVERS WHICH WE WANTS TO MONITOR, IT FIRST GATHER SYSTEM METRICS , LIKE CPU-UTILIZATION, MEMORY UTILIZATION AND ETC

THE DATA IS STORE IN TSDB (TIME SERIES DATABASE) WHICH KEEPS DATA AND ALL THE CHANGES WITH OVER TIME

HTTP SERVER: AS PROMETHEUS ALSO HAVE ITS OWN WEB USER INTEPHASE (WEB UI), SO IT RUNS OVER THE HTTP SERVER

PROMQL: PromQL, short for Prometheus Querying Language, is the main way to query metrics within Prometheus. You can display an expression's return either as a graph or export it using the HTTP API. PromQL uses three data types: scalars, range vectors, and instant vectors.

GRAFANA: YOU CAN ALSO USE GRAFANA TO SEE THE GRAPHS METRICS, THE GRAFANA SUPPORTS PROMQL

API CLIENT: LIKE IF YOU WANT TO SHOW SOME PROMETHEUS DATA IN YOUR CUSTOM WEBSITE SO YOU CAN DO IT BY API-CLIENT BY MAKING API CALLS

ALERT-MANAGER: he Alertmanager handles alerts sent by client applications such as the Prometheus server. It takes care of deduplicating, grouping, and routing them to the correct receiver integration such as email, PagerDuty, or OpsGenie. It also takes care of silencing and inhibition of alerts. So basically whenever a event will happen it will trigger the alertmanager and it will send the notifications and the event we will define in the config file

INSTALLATION PROMETHEUS

- LAUNCH AN EC2 INSTANCE USING LINUX SERVER AMI ND CONNECT IT TO THE GITBASH TERMINAL
- MAKE AN DIRECTORY mkdir prometheus
- cd prometheus
- <https://prometheus.io/download/> GO HERE AND RIGHT CLICK INFRONT OF LINUX AND COPY THE LINK ADDRESS
- NOW COME TO THE TERMINAL GITBASH AND TYPE = # wget AND PASTE THE LINK ADDRESS WE COPIED

```
wget https://github.com/prometheus/prometheus/releases/download/v3.0.1/prometheus-3.0.1.linux-amd64.tar.gz
```

- NOW IF YOU WILL SEE BY = ls IT WILL SHOW THE FILE IS IN THE .tar CONDITION SO UNTAR IT
- # tar xzvf prometheus-2.54.0.linux-amd64.tar.gz

- NOW YOU CAN SEE A DIRECTORY IS MADE LIKE === prometheus-2.54.0.linux-amd64
- GO INSIDE IT # cd prometheus-2.54.0.linux-amd64
- # ls -lrt AND YOU CAN SEE THERE ARE = prometheus , promtool , prometheus.yml file
- GO INSIDE ===]# vi prometheus.yml
- IN THE FILE YOU CAN SEE A -target section IN THERE CHANGE THE LOCALHOST WITH THE PUBLIC IP OF THE SERVER YOU ARE WORKING ON AND ON THIS SERVER THE PROMETHEUS WILL RUN
- AFTER CHANGES RUN THI COMMAND TO START PROMETHEUS ===== ./prometheus --config.file=prometheus.yml
- GO TO THE INSTANCE COPY THE PUBLIC IP AND PASTE ON THE GOOGLE AND :9090 AS PROMETHEUS RUNS ON THE 9090 PROT
- type cpu and first query hit it and click on graph so you can see the monitoring of your server

NODE_EXPORT

- AFTER PROMETHEUS MONITORING ITSELF WE WILL NOW MONITOR ANOTHER SERVER WITH THE HELP OF NODE EXPORTER
- WHAT IS NODE_EXPORTER: A Node Exporter is needed on all servers or virtual machines to collect data on all nodes; it runs on the port 9100 default.

WORKING WITH NODE EXPORTER

- FIRST INSTALL THE PROMETHEUS ON THE MAIN SERVER WHERE THE PROMETHEUS WILL RUN WITH THE ABOVE STEPS AND START THE PROMETHEUS
- THEN TAKE AN ANOTHER SERVER WHERE WE WILL INSTALL THE NODE_EXPORTER SO THAT WE CAN MONITOR THE DATA OF THAT SERVER
- AFTER SUCCESSFULLY CONNECTED TO THE GITBASH OR ANY OTHER TERMINAL FOLLOW BELOW STEPS



- `# yum update`
- TO DOWNLOAD NODE_EXPORTER GO TO THIS PAGE = https://prometheus.io/download/#node_exporter AND COPY THE LINK ADDRESS OF THE LINUX
- `# wget https://github.com/prometheus/node_exporter/releases/download/v1.8.2/node_exporter-1.8.2.linux-amd64.tar.gz`
- THE FILE IS .tar UNTAT IT = `# tar -zxvf node_exporter-1.8.2.linux-amd64.tar.gz`
- GO INSIDE THAT DIRECTORY = `cd node_exporter-1.8.2.linux-amd64/`
- RUN THE NODE EXPORTER BY COMMAND = `# ./node_exporter`
- NOW GO TO THE INSTANCE COPY THE PUBLIC IP OF NODE EXPORTER AND TYPE 9100 CHECK IF IT STARTED OR NOT
- NOW OPEN NEW PROMETHEUS TERMINAL BECOUSE WE WILL CONFIGURE IN THE promtheus.yml file

- `# vi prometheus.yml`
- `- targets: ["3.108.54.19:9100"]` `# ADD SUCH TARGET DOWN TO THE`
`PROMETHEUES TARGET WE GAVE AND PUBLIC IP OF NOE EXPORTER SERVER AND :PORT`
- NOW WE NEED TO RESTART THE PROMETHEUS SO THAT CHANGES SHOULD IMPLEMENT
- GO TO THE PROMETHEUS SERVER STOPP IT AND RUN = `# ./prometheus --`
`config.file=prometheus.yml`
- NOW THE NODE EXPORTER WILL SEND THE DATA TO THE PROMETHEUS
- GO TO THE PROMETHEUS TYPE THIS QUERY = `avg by (job, instance,`
`mode)(rate(node_cpu_seconds_total[5m]))` AND EXECUTE NOW CLICK ON GRAPH AND
YOU CAN SEE THE METRICS
- GO TO THE STATUS AND CLICK ON TARGETS AND THERE YOU CAN SEE BOTH THE SERVERS
AND THEIR STATUS UP.